

11. Local structure of s -dimensional sets and measures

Let E be an $\mathcal{H}^s$ measurable subset of $\mathbf{R}^n$ with $0 < \mathcal{H}^s(E) < \infty$. From the upper density estimates in Theorem 6.2(1) we already know that for $\mathcal{H}^s$ almost all $x \in E$ there are arbitrarily small radii r such that $\mathcal{H}^s(E \cap B(x, r)) \sim r^s$. So we know roughly how much of E there is in such small balls $B(x, r)$ but we would also like to know something about how E is distributed there. The bigger s is the more effectively E should fill $B(x, r)$ in some sense. One way to formulate this is the following special case of the results of this chapter. Let $n - 1 < s \leq n$. Given any $\delta > 0$ there is $c(\delta) > 0$ such that for $\mathcal{H}^s$ almost all $x \in E$ there are arbitrarily small radii r for which

$$\mathcal{H}^s(E \cap B(x, r) \cap \{y : d(y, L) \leq \delta|x - y|\}) \geq c(\delta)r^s$$

for all lines L through x . For $s \leq n - 1$ this is of course no longer true, but as long as $s > m$, $m = 1, \dots, n - 1$, we will be able to say something similar about how much there is E near $(n - m)$ -planes. This chapter is based on Mattila [14].

Distribution of measures with finite energy

More generally, we shall work first with general measures of finite energy. Throughout this chapter m and n will be integers with $0 < m < n$. Let μ be a Radon measure on $\mathbf{R}^n$. We shall use the measures $\mu_{V,a}$ introduced in the preceding chapter. There we only looked at measures $\mu_{V,a}$ parametrized by $V \in G(n, n - m)$ and $a \in V^\perp$, but now we want to introduce also measures $\mu_{V,x}$ on $(n - m)$ -planes $V + x$ through $x \in \mathbf{R}^n$. This is easily done. We simply put

$$\mu_{V,x} = \mu_{V,a} \quad \text{for } x \in P_{V^\perp}^{-1}\{a\}$$

whenever $a \in V^\perp$ is such that $\mu_{V,a}$ is defined. Recall that this holds for $\mathcal{H}^m$ almost all $a \in V^\perp$.

For any $G \subset G(n, n - m)$ and $x \in \mathbf{R}^n$ we shall consider a kind of cone G_x generated by G with vertex at x :

$$(11.1) \quad G_x = \bigcup_{V \in G} V_x \quad \text{where } V_x = V + x.$$

Note that when $m = n - 1$ and G is a ball,

$$G = \{L \in G(n, 1) : L \cap S^{n-1} \cap B(\theta, \delta) \neq \emptyset\}$$

for some $\theta \in S^{n-1}$ and $\delta > 0$, we have as G_x the ordinary two-sided cone with vertex x generated by the spherical cap $S = S^{n-1} \cap B(\theta, \delta)$, that is, $G_x = \{x + ty : y \in S, t \in \mathbf{R}\}$.

Next we derive some information on the measures $\mu_{V,x}$.

11.2. Lemma. Let μ be a Radon measure on $\mathbf{R}^n$ with $I_m(\mu) < \infty$. Then $\mu_{V,x}$ is defined for $\mu \times \gamma_{n,n-m}$ almost all $(x, V) \in \mathbf{R}^n \times G(n, n-m)$. Moreover, for any non-negative lower semicontinuous function g on $\mathbf{R}^n$, the function $(x, V) \mapsto \int g d\mu_{V,x}$ is Borel measurable and there exists a set $E \subset \mathbf{R}^n$, independent of g , such that $\mu(\mathbf{R}^n \setminus E) = 0$ and

$$\iint g d\mu_{V,x} d\gamma_{n,n-m} V \leq c \int g(y) |x - y|^{-m} d\mu y \quad \text{for } x \in E.$$

Here c is a constant depending only on m and n .

Proof. Let D be the countable dense subset of $C_0^+(\mathbf{R}^n)$ used in 10.1. Let Q be the set of the pairs (x, V) for which $\mu_{V,x}$ exists. Then Q consists of those (x, V) such that the finite limit

$$\lim_{\delta \downarrow 0} (2\delta)^{-m} \int_{V_x(\delta)} \varphi d\mu$$

exists for all $\varphi \in D$. This is seen to be a Borel set by standard methods. By 10.1,

$$\mathcal{H}^m(P_{V^\perp} \{x : (x, V) \notin Q\}) = \mathcal{H}^m(\{a \in V^\perp : \mu_{V,a} \text{ is not defined}\}) = 0.$$

This gives $\mu(\{x : (x, V) \notin Q\}) = 0$ whenever $P_{V^\perp} \# \mu \ll \mathcal{H}^m$. By Theorem 9.7 and (3.10), as $I_m(\mu) < \infty$, this holds for $\gamma_{n,n-m}$ almost all $V \in G(n, n-m)$. Hence $\mu_{V,x}$ is defined for $\mu \times \gamma_{n,n-m}$ almost all (x, V) .

For the proof of the rest of the lemma, we may assume that $g \in C_0^+(\mathbf{R}^n)$ by approximating with a non-decreasing sequence of such functions. Due to the definition of $\mu_{V,x}$ and (10.3) we have

$$\int g d\mu_{V,x} = \lim_{\delta \downarrow 0} (2\delta)^{-m} \int_{V_x(\delta)} g d\mu \quad \text{for } (x, V) \in Q.$$

Since Q is a Borel set whose complement has zero $\mu \times \gamma_{n,n-m}$ measure, the Borel measurability of $\int g d\mu_{V,x}$ follows easily from this formula. For E we may take

$$E = \{x : \gamma_{n,n-m}\{V : (x, V) \notin Q\} = 0\}.$$

Then using the above formula, Fatou's lemma, Fubini's theorem, and Lemma 3.11 we estimate for $x \in E$,

$$\begin{aligned} & \iint g d\mu_{V,x} d\gamma_{n,n-m} V \\ & \leq \liminf_{\delta \downarrow 0} (2\delta)^{-m} \iint_{V_x(\delta)} g d\mu d\gamma_{n,n-m} V \\ & = \liminf_{\delta \downarrow 0} (2\delta)^{-m} \int g(y) \gamma_{n,n-m}(\{V : d(y, V_x) \leq \delta\}) d\mu y \\ & \leq c \int g(y) |x - y|^{-m} d\mu y. \end{aligned}$$

□

We derive two corollaries.

11.3. Lemma. Suppose μ is a Radon measure on $\mathbf{R}^n$ with compact support and with $I_m(\mu) < \infty$. Let $x \in E$ where E is as in Lemma 11.2. Then for any Borel set $G \subset G(n, n-m)$,

$$\int_G \int |x-y|^m d\mu_{V,x} y d\gamma_{n,n-m} V \leq c\mu(G_x)$$

where c is the constant of Lemma 11.2.

Proof. Approximating first G from inside with compact sets and then these compact sets from outside with open sets, it is enough to prove the lemma for open sets G . Then also $G_x \setminus \{x\}$ is open. (Actually Lemma 11.2 also holds for Borel functions g , but the approximation is slightly simpler here.) Applying Lemma 11.2 with

$$g(y) = |x-y|^m \chi_{G_x \setminus \{x\}}(y)$$

one obtains

$$\begin{aligned} & \int_G \int |x-y|^m d\mu_{V,x} y d\gamma_{n,n-m} V \\ & \leq \iint g d\mu_{V,x} d\gamma_{n,n-m} V \leq c\mu(G_x). \end{aligned} \quad \square$$

11.4. Lemma. Let μ be a Radon measure on $\mathbf{R}^n$ with compact support. If $m < s < n$, $I_s(\mu) < \infty$, and $\varrho > 0$, then

$$\iint \mu_{V,x}(B(x, \varrho)) d\gamma_{n,n-m} V d\mu x \leq c\varrho^{s-m} I_s(\mu),$$

where c is the constant of Lemma 11.2.

Proof. By approximation it suffices to prove this for the open balls $U(x, \varrho)$ instead of $B(x, \varrho)$. Since μ has compact support, $s > m$ and $I_s(\mu) < \infty$, also $I_m(\mu) < \infty$. Applying Lemma 11.2 to the characteristic functions of the balls $U(x, \varrho)$, $x \in E$, we obtain

$$\begin{aligned} & \iint \mu_{V,x}(U(x, \varrho)) d\gamma_{n,n-m} V d\mu x \\ & \leq c \iint \int_{U(x, \varrho)} |x-y|^{-m} d\mu y d\mu x \\ & = c\varrho^{s-m} \iint \int_{U(x, \varrho)} (|x-y|/\varrho)^{s-m} |x-y|^{-s} d\mu y d\mu x \\ & \leq c\varrho^{s-m} I_s(\mu). \end{aligned} \quad \square$$

In the case $m = n - 1$ we shall in addition to the cones G_x consider the corresponding one-sided cones. We parametrize them using S^{n-1} : for $\theta \in S^{n-1}$ let L_θ be the half-line

$$L_\theta = \{t\theta : 0 \leq t < \infty\}.$$

For $S \subset S^{n-1}$ and $x \in \mathbf{R}^n$ define the one-sided cone S_x generated by S with vertex at x by

$$(11.5) \quad S_x = \bigcup_{\theta \in S} L_\theta + x = \{t\theta + x : 0 \leq t < \infty, \theta \in S\}.$$

Letting $\ell(\theta) = \{t\theta : t \in \mathbf{R}\}$, the measures $\mu_{\ell(\theta),x}$ lead naturally to measures

$$\mu_{\theta,x} = \mu_{\ell(\theta),x} \llcorner (L_\theta + x)$$

on the half-lines $L_\theta + x$.

We shall now prove a theorem on the global distribution of measures with finite energy. Since the statement is somewhat technical, let us contemplate it in the light of an example.

Suppose μ is the Lebesgue measure restricted to the union of p disjoint balls $B_1, \dots, B_p$ in $B(1)$ each of radius r , normalized so that $\mu(\mathbf{R}^n) = 1$. Then $\mu(B_i) = 1/p$ for all i . Instead of assuming $I_s(\mu) < \infty$, let us assume the closely related condition that for some $s > n - 1$

$$\mu(B(x, R)) \leq R^s \quad \text{for } x \in \mathbf{R}^n, R > 0;$$

recall the discussion at the beginning of Chapter 8. This means essentially that no ball of radius R meets more than roughly pR^s balls B_i for $R \geq r$, and we must have $p \geq r^{-s}$ in order that this could hold for $R = r$. Thus the balls B_i are not allowed to concentrate too much in small regions, but they need not be even nearly uniformly distributed inside $B(1)$. In this situation Theorem 11.6 gives the following information: if we fix a small angle γ and $\varepsilon > 0$, we can find $\delta > 0$, independent of p , such that when looking from a randomly selected ball B_i into a sector with opening angle γ we can see with probability at least $1 - \varepsilon$ some other ball B_j , provided $p > 1/\delta$; in fact we can see at least $p\delta$ such balls.

11.6. Theorem. *Let $m < s < n$ and let ε, γ and I be positive numbers. Then there is $\delta = \delta(m, n, s, \varepsilon, \gamma, I) > 0$ such that*

$$\begin{aligned} \mu\{x : \mu(G_x) < \delta \text{ for some Borel set } G \subset G(n, n-m) \\ \text{with } \gamma_{n,n-m}(G) \geq \gamma\} \leq \varepsilon \end{aligned}$$

whenever μ is a Radon measure on $\mathbf{R}^n$ with $\text{spt } \mu \subset B(1)$, $\mu(\mathbf{R}^n) = 1$ and $I_s(\mu) \leq I$.

If $m = n - 1$, we furthermore have

$$\mu\{x : \mu(S_x) < \delta \text{ for some Borel set } S \subset S^{n-1} \text{ with } \mathcal{H}^{n-1}(S) \geq \gamma\} \leq \varepsilon.$$

Proof. Let $B = B(1)$. We may of course assume that $\varepsilon, \gamma \leq 1$ and $I \geq 1$. Letting c be the constant of Lemma 11.2, define positive numbers ζ, ϱ and δ by

$$\begin{aligned}\zeta &= \gamma\varepsilon/2^{m+2}, \\ \varrho^{s-m} &= (\gamma\varepsilon\zeta)/(8cI) = (\gamma\varepsilon)^2/(2^{m+5}cI), \\ \delta &= \varrho^m\gamma\zeta/(4c).\end{aligned}$$

For the Borel set

$$A_1 = \{x : \gamma_{n,n-m}(\{V : \mu_{V,x}(B) \leq \zeta\}) \geq \tfrac{1}{2}\gamma\},$$

we have by Fubini's theorem

$$\begin{aligned}\tfrac{1}{2}\gamma\mu(A_1) &\leq \int \gamma_{n,n-m}(\{V : \mu_{V,x}(B) \leq \zeta\}) d\mu x \\ &= \int \mu(\{x : \mu_{V,x}(B) \leq \zeta\}) d\gamma_{n,n-m}V.\end{aligned}$$

Since $I_s(\mu) < \infty$ and $s > m$, $P_{V^\perp \#} \mu \ll \mathcal{H}^m$ for $\gamma_{n,n-m}$ almost all $V \in G(n, n-m)$ by Theorem 9.7, and for any such V (10.6) yields for the Borel set $C = \{a \in V^\perp \cap B : \mu_{V,a}(B) \leq \zeta\}$,

$$\begin{aligned}(1) \quad \mu(\{x : \mu_{V,x}(B) \leq \zeta\}) &= \mu(P_{V^\perp}^{-1}(C)) = \int_C \mu_{V,a}(B) d\mathcal{H}^m a \\ &\leq \zeta \mathcal{H}^m(V^\perp \cap B) = 2^m \zeta.\end{aligned}$$

Combining this with the earlier inequality for $\mu(A_1)$, we get

$$\mu(A_1) \leq 2^{m+1}\zeta/\gamma = \varepsilon/2.$$

Next we consider the Borel set

$$A_2 = \left\{x : \int \mu_{V,x}(B(x, \varrho)) d\gamma_{n,n-m}V \geq \gamma\zeta/4\right\}.$$

Applying Lemma 11.4, we obtain

$$\begin{aligned}\mu(A_2) &\leq (\gamma\zeta/4)^{-1} \iint \mu_{V,x}(B(x, \varrho)) d\gamma_{n,n-m} V d\mu x \\ &\leq (\gamma\zeta/4)^{-1} c\varrho^{s-m} I_s(\mu) \leq \varepsilon/2.\end{aligned}$$

Thus $\mu(A_1 \cup A_2) \leq \varepsilon$, and we shall show that, with E as in Lemma 11.2, for $x \in E \setminus (A_1 \cup A_2)$ we have $\mu G_x \geq \delta$ whenever $G \subset G(n, n-m)$ is a Borel set with $\gamma_{n,n-m}(G) \geq \gamma$. By Lemma 11.3,

$$\begin{aligned}c\mu(G_x) &\geq \int_G \int |x-y|^m d\mu_{V,x} y d\gamma_{n,n-m} V \\ &\geq \varrho^m \int_G \mu_{V,x}(B \setminus B(x, \varrho)) d\gamma_{n,n-m} V \\ &\geq \varrho^m \int_G \mu_{V,x}(B) d\gamma_{n,n-m} V - \varrho^m \int \mu_{V,x}(B(x, \varrho)) d\gamma_{n,n-m} V.\end{aligned}$$

Since $x \notin A_1$,

$$\gamma_{n,n-m}(\{V \in G : \mu_{V,x}(B) > \zeta\}) \geq \gamma_{n,n-m}(G) - \gamma/2 \geq \gamma/2,$$

whence

$$\varrho^m \int_G \mu_{V,x}(B) d\gamma_{n,n-m} V \geq \varrho^m \zeta \gamma/2.$$

Since $x \notin A_2$,

$$\varrho^m \int \mu_{V,x}(B(x, \varrho)) d\gamma_{n,n-m} V \leq \varrho^m \gamma \zeta/4.$$

Combining these we get

$$c\mu(G_x) \geq \varrho^m \gamma \zeta/2 - \varrho^m \gamma \zeta/4 = c\delta$$

as required.

To prove the last statement when $m = n-1$, we use the measures $\mu_{\theta,x}$, $\theta \in S^{n-1}$, introduced above. Since $\mu_{\theta,x} \leq \mu_{\ell(\theta),x}$, the proof runs as above if we are able to verify the inequality (1) for the measures $\mu_{\theta,x}$. Note that also Lemma 11.3 holds in the form

$$\int_S \int |x-y|^{n-1} d\mu_{\theta,x} y d\mathcal{H}^{n-1} \theta \leq c\mu(S_x).$$

Set

$$B_\theta = \{x : \mu_{\theta,x}(B) \leq \zeta\}.$$

For each $a \in \ell(\theta)^\perp$ for which $\mu_{\ell(\theta),a}$ is defined, $B_\theta \cap \ell(\theta)_a$ is either a half-line in direction θ or the whole line $\ell(\theta)_a$ depending on whether $\mu_{\ell(\theta),a}(B) > \zeta$ or $\mu_{\ell(\theta),a}(B) \leq \zeta$. In both cases $\mu_{\ell(\theta),a}(B_\theta) \leq \zeta$. Thus for $\mathcal{H}^{n-1}$ almost all $\theta \in S^{n-1}$,

$$\mu(B_\theta) = \int_{\ell(\theta)^\perp} \mu_{\ell(\theta),a}(B_\theta) d\mathcal{H}^{n-1}a \leq \zeta \mathcal{H}^{n-1}(\ell(\theta)^\perp \cap B). \quad \square$$

Conical densities

As an application of the previous theorem we shall derive a result on the following upper densities for Hausdorff measures.

11.7. Definition. Let $0 \leq s \leq n$, $A \subset \mathbf{R}^n$ and $x \in \mathbf{R}^n$. Define for $\gamma > 0$,

$$\Theta_m^{*s}(\gamma, A, x) = \limsup_{r \downarrow 0} (2r)^{-s} \Theta_m^s(\gamma, A, x, r)$$

where

$$\Theta_m^s(\gamma, A, x, r) = \inf \left\{ \mathcal{H}^s(A \cap G_x \cap B(x, r)) : G \subset G(n, n-m) \right. \\ \left. \text{is a Borel set with } \gamma_{n,n-m}(G) \geq \gamma \right\}.$$

If $m = n - 1$, we define the corresponding one-sided upper density by

$$\tilde{\Theta}^{*s}(\gamma, A, x) = \limsup_{r \downarrow 0} (2r)^{-s} \tilde{\Theta}^s(\gamma, A, x, r)$$

where

$$\tilde{\Theta}^s(\gamma, A, x, r) = \inf \left\{ \mathcal{H}^s(A \cap S_x \cap B(x, r)) : S \subset S^{n-1} \right. \\ \left. \text{is a Borel set with } \mathcal{H}^{n-1}(S) \geq \gamma \right\}.$$

11.8. Theorem. Let $m < s < n$, $\gamma > 0$ and $c = c(m, n, s, \gamma) = 12^{-s} \delta$ where $\delta = \delta(m, n, s, \frac{1}{2}, \gamma, 3s12^s/(s-m))$ is defined as in Theorem 11.6. If $A \subset \mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$, then

$$\Theta_m^{*s}(\gamma, A, x) \geq c \quad \text{for } \mathcal{H}^s \text{ almost all } x \in A.$$

If $m = n - 1$, we furthermore have

$$\tilde{\Theta}^{*s}(\gamma, A, x) \geq c \quad \text{for } \mathcal{H}^s \text{ almost all } x \in A.$$

Proof. With the help of Exercise 1.2 we may assume that A is $\mathcal{H}^s$ measurable. If the first statement is false, we may assume, replacing A with a suitable subset, that $\mathcal{H}^s(A) > 0$ and $\Theta_m^{*s}(\gamma, A, x) < c$ for $x \in A$. Replacing A with another subset, we may also assume that there is $r_0 > 0$ such that

$$(1) \quad \Theta_m^s(\gamma, A, x, r) < c3^s r^s \quad \text{for } x \in A, 0 < r < r_0,$$

and by Theorem 6.2 (1),

$$\mathcal{H}^s(A \cap B(x, r)) < 6^s r^s \quad \text{for } x \in \mathbf{R}^n, 0 < r < r_0,$$

(recall the brief argument before 8.3 to get this for all $x \in \mathbf{R}^n$). By the other inequality in 6.2 (1) we find and fix $x \in A$ and r such that $0 < r < r_0/2$ and

$$\mathcal{H}^s(A \cap B(x, r)) \geq 2^{-s} r^s.$$

Define a Radon measure μ by the formula

$$\mu(E) = \mathcal{H}^s(\{x + ry : y \in E\} \cap A \cap B(x, r)) / \mathcal{H}^s(A \cap B(x, r)).$$

This means that μ is the normalized image of $\mathcal{H}^s \llcorner (A \cap B(x, r))$ under the transformation $y \mapsto (y - x)/r$ sending the ball $B(x, r)$ to $B(1)$. We then have $\text{spt } \mu \subset B(1)$ and $\mu(\mathbf{R}^n) = 1$. For $z \in \mathbf{R}^n$ and $0 < \varrho \leq 1$,

$$\begin{aligned} \mu(B(z, \varrho)) &= \mathcal{H}^s(B(x + rz, r\varrho) \cap A \cap B(x, r)) / \mathcal{H}^s(A \cap B(x, r)) \\ &\leq 6^s (r\varrho)^s / (2^{-s} r^s) = 12^s \varrho^s. \end{aligned}$$

Let $t = (s + m)/2$. Computing as early in Chapter 8 we obtain for $z \in \mathbf{R}^n$,

$$\begin{aligned} \int |y - z|^{-t} d\mu y &= \int_0^\infty \mu(B(z, u^{-1/t})) du \\ &\leq \mu(\mathbf{R}^n) + \int_1^\infty \mu(B(z, u^{-1/t})) du \\ &\leq 1 + 12^s \int_1^\infty u^{-s/t} du = 1 + 12^s t / (s - t) \leq 3s 12^s / (s - m). \end{aligned}$$

Hence $I_t(\mu) \leq 3s12^s/(s-m)$. Theorem 11.6 now tells us that for a set of points $z \in B(0, 1)$ having a positive μ measure, $\mu(G_z) \geq \delta$ for all Borel sets $G \subset G(n, n-m)$ with $\gamma_{n, n-m}(G) \geq \gamma$. For some such z the point $y = x + rz$ belongs to $A \cap B(x, r)$ by the definition of μ . Then for any Borel set $G \subset G(n, n-m)$ with $\gamma_{n, n-m}(G) \geq \gamma$,

$$\begin{aligned}\mathcal{H}^s(A \cap B(y, 2r) \cap G_y) &\geq \mathcal{H}^s(A \cap B(x, r) \cap G_y) \\ &= \mu(G_z) \mathcal{H}^s(A \cap B(x, r)) \geq \delta 2^{-s} r^s = c 6^s r^s,\end{aligned}$$

whence $\Theta_m^s(\gamma, A, y, 2r) \geq c 6^s r^s$, which contradicts (1).

This proves the first statement. The second statement follows similarly with the help of the second statement in Theorem 11.6. $\square$

We now derive as consequences of Theorem 11.8 some angular density and porosity theorems. For the first we use the following notation.

11.9. Definition. For $V \in G(n, n-m)$, $a \in \mathbf{R}^n$, $0 < \alpha < 1$ and $0 < r \leq \infty$, we define

$$X(a, r, V, \alpha) = \{x : d(x - a, V) < \alpha|x - a|, |x - a| \leq r\}.$$

This is a kind of cone around the plane $V + a$. The number α measures the opening angle of this cone; in the case $n - m = 1$ it is the sine of this angle.

These cones are special cases of the cones G_x we considered above. It is a simple exercise in linear algebra to show that for any $V \in G(n, n-m)$ and $\alpha \in (0, 1)$ there is an open set $G \subset G(n, n-m)$ such that $X(a, \infty, V, \alpha) = G_a \setminus \{a\}$ for $a \in \mathbf{R}^n$. Obviously the measure of G depends only on α . In the case $m = n - 1$, G is a ball on $G(n, 1)$. These observations show that the following theorem is a weaker form of Theorem 11.8.

11.10. Theorem. Let $V \in G(n, n-m)$, $m < s \leq n$, $0 < \alpha < 1$, and let A be a subset of $\mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$. Then for $\mathcal{H}^s$ almost all $x \in A$

$$(1) \quad \limsup_{r \downarrow 0} r^{-s} \mathcal{H}^s(A \cap X(x, r, V, \alpha)) \geq c$$

where c is a positive constant depending only on m, n, s and α . In the case $n - 1 < s \leq n$ we have also for $\theta \in S^{n-1}$,

$$(2) \quad \limsup_{r \downarrow 0} r^{-s} \mathcal{H}^s(A \cap X^+(x, r, \theta, \alpha)) \geq c,$$

where $X^+(x, r, \theta, \alpha) = S_x$ is the one-sided cone as in (11.5) corresponding to the spherical cap $S = S^{n-1} \cap B(\theta, \alpha)$.

We can read from the proofs explicit expressions for the constants in Theorems 11.6, 11.8 and 11.10. They may not be very bad in Theorems 11.6 and 11.8, although the best constants are not known. But the constant c in Theorem 11.10 is certainly far from optimal when $s - m$ is small. From the above we would get something like $\eta^{1/\eta} \alpha^{1/\eta}$ where $\eta \approx s - m$ when the best possible constant c in 11.10(1) is of the form $c(n, m) \alpha^m$. In this form Theorem 11.10 was first proved by Marstrand [1] for $n = 2$, $m = 1$, and then generalized by Salli [1] to arbitrary dimensions. In the case $m = n - 1$ Marstrand and Salli proved their results with constant $c(n)(s - n + 1) \alpha^{n-1}$ for the one-sided cones of 11.10(2). Moreover, Salli had similar results for very general cones.

One cannot replace in Theorem 11.10 the conical upper density by the corresponding lower density even if A had positive lower density $\Theta_*^s(A, x)$ for all $x \in A$. In fact, the lower density must be zero very often according to the following result of Marstrand [1] in $\mathbf{R}^2$. For the formulation of a higher dimensional version, let

$$H(a, \theta) = \{x \in \mathbf{R}^n : (x - a) \cdot \theta > 0\}$$

be the half-space containing $a + \theta$ whose boundary contains a and is orthogonal to θ . For $\eta > 0$, let

$$H(a, \theta, \eta) = \{x \in \mathbf{R}^n : (x - a) \cdot \theta > \eta |x - a|\}.$$

11.11. Theorem. Let $0 < s < n$ and let A be an $\mathcal{H}^s$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$.

(1) If $\eta > 0$, then for $\mathcal{H}^s$ almost all $x \in A$, $\Theta_*^s(A \cap H(x, \theta, \eta), x) = 0$ for some $\theta \in S^{n-1}$.

(2) If $n-1 < s < n$, then for $\mathcal{H}^s$ almost all $x \in A$, $\Theta_*^s(A \cap H(x, \theta), x) = 0$ for some $\theta \in S^{n-1}$.

(3) If $0 < s < 1$, then for every $\theta \in S^{n-1}$, $\Theta_*^s(A \cap H(x, \theta), x) = 0$ for $\mathcal{H}^s$ almost all $x \in A$.

We shall give a proof for (1) and (2) in Chapter 14 using tangent measures which will be introduced there. Marstrand's argument from Marstrand [1, pp. 295–297] could also be generalized, see also Falconer [4, 4.7]. We omit the proof of (3). It can be verified with an argument similar to that in Marstrand [1, pp. 293–294], see also Falconer [4, pp. 56–57].

Porosity and Hausdorff dimension

As another application of Theorem 11.8 we shall show that very strongly porous sets in $\mathbf{R}^n$ can have Hausdorff dimension only slightly above $n - 1$. First we give some notation.

11.12. Definition. For $A \subset \mathbf{R}^n$, $x \in \mathbf{R}^n$ and $r > 0$, we set

$$p(A, x, r) = \sup \{ \varrho : B(z, \varrho) \subset B(x, r) \setminus A \text{ for some } z \in \mathbf{R}^n \}.$$

The (strong) porosity of A at x is defined as

$$p(A, x) = \liminf_{r \downarrow 0} p(A, x, r)/r.$$

11.13. Remarks. Clearly for $x \in A$, $p(A, x, r)$ is between 0 and $r/2$ and $p(A, x)$ between 0 and $1/2$.

It is fairly easy to see by a cubical division argument (see e.g. Salli [2] for a discussion on this) that there is $d(p) < n$ for $0 < p < 1/2$ such that $\dim A \leq d(p)$ whenever $A \subset \mathbf{R}^n$ and $p(A, x) \geq p > 0$ for $x \in A$. In the next theorem we show that $d(p)$ can be chosen to tend to $n - 1$ as $p \uparrow \frac{1}{2}$.

11.14. Theorem. For $0 < p < 1/2$ there is $d(p)$, $n - 1 \leq d(p) \leq n$, such that

$$(1) \quad \lim_{p \uparrow \frac{1}{2}} d(p) = n - 1$$

and $\dim A \leq d(p)$ whenever $A \subset \mathbf{R}^n$ is such that $p(A, x) \geq p$ for all $x \in A$.

Proof. Let $d(p)$ be the smallest d such that $\dim A \leq d$ whenever $A \subset \mathbf{R}^n$ is such that $p(A, x) \geq p$ for all $x \in A$. Obviously it exists. We should show that (1) holds. This means that for $s > n - 1$ there is $p(s) < 1/2$ such that if $d(p) \geq s$ then $p \leq p(s)$. We shall verify the following equivalent statement.

For each $s > n - 1$ there is $p(s) < 1/2$ such that if there exist $p > 0$ and $A \subset \mathbf{R}^n$ with $\dim A > s$ and $p(A, x) > p$ for $x \in A$, then $p \leq p(s)$.

Suppose A satisfies these conditions. Then A has a subset B with $\dim B > s$ and $p(B, x, r) \geq pr$ for $x \in B$, $0 < r < r_0$, for some $r_0 > 0$. Clearly the closure of B , say C , has the same property. As $\mathcal{H}^s(C) = \infty$,

C has by Theorem 8.19 a closed subset F such that $0 < \mathcal{H}^s(F) < \infty$. For F we still have

$$(2) \quad p(F, x, r) \geq pr \quad \text{for } x \in F, \ 0 < r < r_0.$$

We may assume $1/3 < p \leq 1/2$. Let $1/3 < q < p$. If $B(z, qr) \subset B(x, r) \subset \mathbf{R}^n$, $z \neq 0$ and $S = S^{n-1} \cap B((z-x)/|z-x|, 1-2q)$, then (recall (11.5) for the notation)

$$(3) \quad S_x \cap B(x, r/4) \subset B(x, 2(1-2q)r) \cup B(z, qr).$$

We leave the simple verification to the reader. Setting $\gamma = \mathcal{H}^{n-1}(S^{n-1} \cap B(\theta, 1-2q))$ for $\theta \in S^{n-1}$, let $c = c(n-1, n, s, \gamma)$ be the constant of Theorem 11.8. Using Theorem 6.2(1) and Theorem 11.8 we find $x \in F$ and $r < r_0$ such that

$$(4) \quad \mathcal{H}^s(F \cap B(x, 2(1-2q)r)) \leq 5^s(1-2q)^s r^s$$

and

$$(5) \quad \mathcal{H}^s(F \cap S'_x \cap B(x, r/4)) \geq c3^{-s}r^s$$

whenever $S' \subset S^{n-1}$ is a Borel set with $\mathcal{H}^{n-1}(S') \geq \gamma$. Because of (2) we can choose z so that $B(z, qr) \subset B(x, r) \setminus F$. Applying (5) to the spherical cap S as in (3) and using (4), we then have

$$\begin{aligned} c3^{-s}r^s &\leq \mathcal{H}^s(F \cap S_x \cap B(x, r/4)) \\ &\leq \mathcal{H}^s(F \cap [B(x, 2(1-2q)r) \cup B(z, qr)]) \\ &= \mathcal{H}^s(F \cap B(x, 2(1-2q)r)) \leq 5^s(1-2q)^s r^s, \end{aligned}$$

whence $q \leq \frac{1}{2} - c^{1/s}/30$. Since q was chosen arbitrarily between $1/3$ and p , we have finally $p \leq \frac{1}{2} - c^{1/s}/30$. $\square$

11.15. Remarks. Salli [2] gave a different more direct proof for Theorem 11.14. It also applies to the Minkowski dimension; but then one has to assume the uniform inequality

$$p(A, x, r) \geq pr \quad \text{for } x \in A, \ 0 < r < r_0.$$

Salli even derived the optimal behaviour of $d(p)$ when $p \uparrow \frac{1}{2}$. He showed that

$$d(p) = n - 1 + \frac{c}{\log(1/(1-2p))}$$

suffices, where c depends only on n , and that with some other constant b depending only on n there exists for all p , $0 < p < 1/2$, $A_p \subset \mathbf{R}^n$ such that $p(A_p, x) \geq p$ for $x \in A_p$ and $\dim A_p \geq n - 1 + b/\log(1/(1 - 2p))$. In $\mathbf{R}^1$, the Cantor sets $C(\lambda)$ of 4.10 serve as examples for the extremal case. In higher dimensions one can use $C(\lambda) \times \mathbf{R}^{n-1}$.

Recently porous sets have received much attention in connection with several problems of analysis. Sets satisfying the porosity condition of Theorem 11.14 have been studied for example in Väisälä [1]. An even greater rôle in analysis has been played by the sets A which are porous in the weaker sense so that for $x \in A$

$$\limsup_{r \downarrow 0} \frac{p(A, x, r)}{r} \geq p.$$

They were first considered in the works of Denjoy [1] and Dolzenko [2]. The survey of Zajíček [1] is a good source for recent references and results. However, there is not much one can say about their Hausdorff dimension. Even when $p = 1/2$ such a set A can have Hausdorff dimension n ; recall the very porous sets of large Hausdorff dimension at the end of 4.12.

Exercises.

Let $A = C(\lambda) \times C(\lambda) \subset \mathbf{R}^2$, $0 < \lambda < 1/2$, where $C(\lambda)$ is the Cantor set of 4.10.

1. Let $\lambda > 1/4$, that is, $\dim A > 1$. Verify the conclusions of Theorem 11.10 without relying on the general theory.
2. Let $\lambda < 1/4$. Show that there exist $L \in G(2, 1)$ and $\alpha > 0$ such that $A \cap X(x, \infty, L, \alpha) = \emptyset$ for all $x \in A$. (Recall Exercise 9.4.)
3. Try to see what happens when $\lambda = 1/4$. We come to this more generally in Chapter 15.
4. Estimate the porosity $p(A, x, r)$.
5. Show that there exists $\theta \in S^1$ such that $\Theta_*^s(A \cap H(x, \theta)) = 0$ for $\mathcal{H}^s$ almost all $x \in A$ where $s = \dim A$.